SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 **Total Marks:50**

Annexure A

Constraint - Based Problem-Solving **Duration: 2Hrs**

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.

   (10 Marks)

2. Implement an ε-greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ε values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.

   (10 Marks)

3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints.

   (10 Marks)

4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.

   (10 Marks)

5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.

(10 Marks)

6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.

(10 Marks)

7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth. Explain how bandwidth constraints influence the reward function and policy optimization.

(10 Marks)

8. Analyse an experimental comparison between random action selection and ε-greedy action selection in a Multi-Armed Bandit problem under a fixed time constraint. Record cumulative rewards and explain which strategy provides better performance.

(10 Marks)

9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.

(10 Marks)

10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

(10 Marks)

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Grid word Reinforcement Learning

Problem understanding: The Objective is to design a Reinforcem-ement Learning agent that learns to Aggriate a grid world environment. The agent must move from that Start position to the goal while avoiding obstacles & taking the Shortest path.

Constraint analysis:

* The agent cannot move outside the grid boundaries
* The agent must avoid obstacle cells.
* Only four movements are allowed up down, left & Right
* The goal must be reached with minimum steps.

Solution Development:

The Grid world environments is created using open AI Gym. A Q-learning algorithm is used to train the agent During training the agent recieves rewards based on its actions. & gradually lessons the optimal path to the goal.

Applications:

* States Spaces: Every cell in the grid represents a state
* actions Spaces: up, down, left, right
* Reward function

Goal : +10

obstacle : -10

Normal : -1

Justifications:

Reinforcement learning allows the agent to learn through -trail & error. The reward functions encourage the agent to rea -ch the goal quickey while avoiding obstacles.

(2) E- Greedy MultiArmed Bandit:

Problem understanding: The objective is to maximize the total reward by selection the best arm (option) within a limited budget of 500 trains.

Constraint analysis:

- Maximum of no. of trails is 500
- The algorithm must became exploration & exploation.

Solution Development:

The e greedy algorithm selects a random arm with probability & and the best known arm with probability the experiment is repeated is for & values of 0.1, 0.3, 0.5

Applications:

E-value	exploration	explealation	Performance
0.1	low	High	High reward after learning
0.2	Medium	Medium	Balanced performance
0.5	High	low	More exploration lower final reward

Justification:

- ε goal 0.1 gives better long term rewards because it is mostly exploits best arm
- ε = 0.3 provides a balances low exploration & exploitation
- ε = 0.5 explain frequently but may reduce overall rewards due to exercise exploration.

3 Markov Decision process (MDP)

Problem understanding:

- Robot must reach destination
- Minimize energy usage
- Choose the best actions

Constraints analysis:

- Battery energy limited
- Robot should avoid unessary movement
- Robot must reach the destination safely

Solution Development:

The problem is modeled as a markov Decision process. The robot selects actions based on current states to maxi-mize rewards while minimizing energy usage.

Justification:

The map helps robot choose actions that minimize energy consumption while ensuring it reaches the destination efficiency.

4. Value Iteration Algorithm:

Problem understanding:

The objective is to determine optimal policy for reaching goal while avoiding restricted areas.

Constraints analysis:

- Restricted cells cannot be entered
- The agent must remain inside grid.
- Shortcuts & safest pain should be selected.

Solution Development:

Value iteration algorithm repeated updates the values of every state until convergance policy is then obtained by selecting action with highest value.

Bellaman Eqn: $V(s) = \max[R(s,a) + V(s')]$

Justification:

value iteration guarames optical policy by repeating applying Bellom Equation. It helps agent avoid restricted states & maximize future words. rewards.

5. Warehouse Robot Battery Constraints:

- Robot makes deliveries
- Battery lasts only 30min
- maximize completed delivries.

Constraint analysis:

- Battery limit re-fixed
- avoid unnessacery travel
- Return before battery ends.

Solution Development:

- Design RL-frame work
- Learn efficient outes
- optimize delivery sequence

Application of Concepts:

- State : Robot position & battery level
- Policy: choose shortest path
- Rewards: +10 per delivery, −5 for battery work

Justification:

- RL learns efficient delivery strategies
- Battery Constraints is satisfied.